

The Exceptional Series of Lie Groups

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Translator's note. This is a direct English working translation from the project TeX transcription. Mathematical notation has been kept close to the source; the prose has been normalized into contemporary English.

Abstract. Numerology of exceptional Lie groups and a conjectural explanation.

Let G_0 be the split adjoint group of one of the following types:

$$A_1, A_2, G_2, D_4, F_4, E_6, E_7, E_8.$$

Fix a pinning of G_0 . Let Γ denote the group of automorphisms of G_0 . For Γ the group of automorphisms of the Dynkin diagram, identified with the group of automorphisms of G_0 preserving the pinning, this is a semidirect product $\Gamma \ltimes G_0$. The group G is an algebraic group over $\mathbb{Q}$.

It would be enough for our purposes to consider the complex Lie group $G(\mathbb{C})$, or even a compact form of $G(\mathbb{C})$. The possibility, for our purpose, of including D_4 in the list was discovered by Cohen and de Man.

Let T be the maximal torus of G supplied by the pinning, and let $\langle \cdot, \cdot \rangle$ be the canonical bilinear form on $\text{Lie}(T)^\vee$: the inverse form to the restriction to $\text{Lie}(T)$ of the Killing form $\text{Tr}(\text{ad } x \text{ ad } y)$. Let α be the highest root and set $k = \langle \alpha, \alpha \rangle$. The values of k are

$$1/2, \quad 1/3, \quad 1/4, \quad 1/6, \quad 1/9, \quad 1/12, \quad 1/18, \quad 1/30.$$

Finally let h be the Coxeter number, equal either to k or to $-(1/6) - k$. Put $a := 1/h$. To this list we also add the trivial group, with $a = 5/6$ or $a = -1$.

Let $\mathfrak{g}$ be the Lie algebra of G , let U be its enveloping algebra, and let $C \in U$ be the Casimir element corresponding to the Killing form. If V is a representation of G , we will call the action of C the Casimir endomorphism of V . If V is irreducible, this is multiplication by a scalar, called the Casimir of V . Recall that the Casimir of the adjoint representation is 1.

Our observation is that, for (G, a) as above, G has virtual representations X_i ($0 \leq i \leq 4$), Y_i ($0 \leq i \leq 3$), Y_i^* ($0 \leq i \leq 3$), A , C , and C^* with the following properties (A)–(G).

(A) Each of these representations is either irreducible, zero, or the negative of an irreducible representation. One has $X_0 = Y_0 = Y_0^* = 1$, the trivial representation, and $X_1 = Y_1 = Y_1^* = \mathfrak{g}$, the adjoint representation.

For each representation R of the symmetric group S_n , there is a functor

$$V \longmapsto [R](V) := \text{Hom}_{S_n}(R, V^{\otimes n}).$$

If V is a representation of a group, the class of $[R](V)$ in the Grothendieck group of representations is a λ -polynomial, that is, a polynomial in the $\wedge^i V$, in the class of V . If R is irreducible and corresponds to a partition p of n , we write $[p]_V$ for the class $[R](V)$. For example,

$$[(n)]_V = \text{Sym}^n V, \quad [(1, \dots, 1)]_V = \wedge^n V, \quad [(2, 1)]_V = V \otimes \wedge^2 V - \wedge^3 V,$$

$$[(3)]_V = V^3 - 2V \cdot \wedge^2 V + \wedge^3 V.$$

(B) In the λ -ring of representations of G , one has

$$\begin{aligned} \text{(B1)} \quad & [(2)]_{\mathfrak{g}} = \text{Sym}^2 \mathfrak{g} = 1 + Y_2 + Y_2^*, \\ \text{(B2)} \quad & [(1, 1)]_{\mathfrak{g}} = \wedge^2 \mathfrak{g} = \mathfrak{g} + X_2, \\ \text{(B3)} \quad & [(3)]_{\mathfrak{g}} = \text{Sym}^3 \mathfrak{g} = \mathfrak{g} + X_2 + A + Y_3 + Y_3^*, \\ \text{(B4)} \quad & [(2, 1)]_{\mathfrak{g}} = 2\mathfrak{g} + X_2 + Y_2 + Y_2^* + A + C + C^*, \\ \text{(B5)} \quad & [(1, 1, 1)]_{\mathfrak{g}} = \wedge^3 \mathfrak{g} = 1 + X_2 + Y_2 + Y_2^* + X_3, \\ \text{(B6)} \quad & Y_2 \otimes \mathfrak{g} = \mathfrak{g} + X_2 + Y_2 + A + Y_3 + C, \\ \text{(B7)} \quad & Y_2^* \otimes \mathfrak{g} = \mathfrak{g} + X_2 + Y_2^* + A + Y_3^* + C^*. \end{aligned}$$

These formulae are permuted by the involution $*$, provided that A and the X_i —in particular $\mathfrak{g}$ —are declared fixed by $*$.

(C) As a consequence of (B), in the λ -ring of representations of G one has

$$\begin{aligned} \text{(C1)} \quad & X_2 = \wedge^2 \mathfrak{g} - \mathfrak{g}, \\ \text{(C2)} \quad & X_3 = \wedge^3 \mathfrak{g} - \mathfrak{g} \cdot \wedge^2 \mathfrak{g} + \text{Sym}^2 \mathfrak{g} + \mathfrak{g} - 1. \end{aligned}$$

Moreover,

$$X_4 = \wedge^4 \mathfrak{g} - \mathfrak{g} \cdot \wedge^3 \mathfrak{g} + \text{Sym}^2 \mathfrak{g} \cdot \wedge^2 \mathfrak{g} - \wedge^2 \mathfrak{g} - (\mathfrak{g} + A) \cdot \mathfrak{g} + \mathfrak{g}.$$

(D) The Casimir has the following values on these representations:

n	X_n	Y_n	Y_n^*	A	C	C^*
n	$n + (n^2 - n)a$	$n + (n^2 - n)a^*$	$8/3$	$3 + 3a$	$3 + 3a^*$	

When the representation in question is zero, this assertion is to be interpreted as empty.

(E) The dimensions of these representations are given by the following formulae, where $\lambda := -6a$:

$$\begin{aligned} \dim X_1 &= -2(\lambda + 5)(\lambda - 6)/\lambda(\lambda - 1), \\ \dim X_2 &= 5(\lambda + 3)(\lambda + 5)(\lambda - 4)(\lambda - 6)/\lambda^2(\lambda - 1)^2, \\ \dim X_3 &= -10(\lambda + 2)(\lambda + 4)(\lambda + 5)(\lambda - 3)(\lambda - 5)(\lambda - 6)/\lambda^3(\lambda - 1)^3, \\ \dim X_4 &= 5(\lambda + 3)(\lambda + 4)(\lambda + 5)(\lambda + 3)(2\lambda - 4)(\lambda - 5)(\lambda - 6)(2\lambda - 5)/\lambda^4(\lambda - 1)^4, \\ \dim Y_2 &= -90(\lambda + 5)(\lambda - 4)/\lambda^2(\lambda - 1)(2\lambda - 1), \\ \dim Y_3 &= -10(\lambda + 5)(5\lambda - 6)(\lambda - 4)(\lambda - 5)(\lambda - 6)/\lambda^3(\lambda - 1)^2(2\lambda - 1)(3\lambda - 1), \\ \dim A &= -27(\lambda + 3)(\lambda + 4)(\lambda + 5)(\lambda - 4)(\lambda - 5)(\lambda - 6)/\lambda^2(\lambda - 1)^2(3\lambda - 1)(3\lambda - 2), \\ \dim C &= 640(\lambda + 3)(\lambda + 5)(\lambda - 3)(\lambda - 5)/\lambda^3(\lambda - 1)(2\lambda - 1)(3\lambda - 2). \end{aligned}$$

For Y_n^* and C^* , the dimension is given by the same formula as for Y_n and C , but with λ replaced by $\lambda^* := 1 - \lambda = -6a^*$. Note that the formulae for the X_i and for A are invariant under $\lambda \mapsto \lambda^*$.

(F) For each of the equalities (B1)–(B7), (C1), and (C2), the formulae (E) express the dimension of the left-hand side, respectively of the right-hand side, as the value at λ of a rational function R_1 , respectively R_2 . One has $R_1 = R_2$ as rational functions.

(G) For each of the equalities (B1)–(B7), (C1), and (C2), the formulae (D), (E), and (1), (2) express the trace of the Casimir on the left-hand side, respectively on the right-hand side, as the value at λ of a rational function R_1 , respectively R_2 . One has $R_1 = R_2$.

If R is a representation of S_n and V is a representation of a semisimple group, the trace of the Casimir endomorphism of $[R](V)$ is obtained from the trace of the Casimir endomorphism of V by the following formula. If χ is the character of R , $n(\sigma)$ is the number of cycles of a permutation $\sigma \in S_n$, and $m(\sigma)$ is the sum of the squares of the cycle lengths of σ , then

$$\mathrm{Tr}(C, [R](V)) = \frac{1}{n!} \sum_{\sigma \in S_n} \chi(\sigma) m(\sigma) (\dim V)^{n(\sigma)-2} \mathrm{Tr}(C, V). \quad (1)$$

For two representations V and W ,

$$\mathrm{Tr}(C, V \otimes W) = \mathrm{Tr}(C, V) \dim(W) + \dim(V) \mathrm{Tr}(C, W). \quad (2)$$

I do not know a satisfactory proof of the preceding assertions. The assertions (B), (C), and (D) are obtained by the brute force of the LiE program. That the dimensions and Casimirs of $\mathfrak{g}$, Y_2 , and Y_2^* have the form (D), (E) for the appropriate a follows in Vogel [2], §6.9, from the fact that the only invariant elements of $\mathrm{Sym}^4(\mathfrak{g})$ are multiples of the square of the invariant element of $\mathrm{Sym}^2(\mathfrak{g})$. The dimension of the X_i follows from that of $\mathfrak{g}$ by (C). If $n = k$, then Y_n is the representation of dominant weight $n\alpha$, with α the highest root, and (D) for Y_n can be deduced from this. The assertion (D) for X_n can be deduced from (C): on the right-hand side, the trace of the Casimir endomorphism is computed by (1).

Assume (D) and the dimension formulae for Y_2 , Y_2^* , and the X_i . If one expresses equality of dimensions and of Casimir traces for both sides of (B3), (B4), (B6), and (B7), one obtains a system of linear equations in $\dim Y_3$, $\dim Y_3^*$, $\dim A$, $\dim C$, and $\dim C^*$, whose right-hand coefficients are rational functions of λ . This gives an expression for these dimensions as rational functions of λ . We used this method to find a polynomial in λ which is a multiple of the denominator, and to estimate the degree of the numerator; the numerator was then obtained by interpolation from the known values corresponding to the pairs (G, a) . That the resulting rational functions factor as in (E) is a miracle. The identities $R_1 = R_2$ of (F), (G) follow simply from the fact that, for a suitable polynomial P in λ , $R_1 P$ and $R_2 P$ are two polynomials in λ which are equal at a number of values of λ strictly greater than their degree.

The conjecture

I conjecture that there exists a semisimple abelian $\mathbb{Q}(t)$ -linear category $\mathcal{C}_t$, equipped with a tensor product $\otimes$ and associativity, commutativity, and unit constraints in the sense of Saavedra [1], Chapter 1, having the following properties.

- (i) The groups $\mathrm{Hom}(X, Y)$ are finite-dimensional over $\mathbb{Q}(t)$.
- (ii) Every object has a dual: $\mathcal{C}_t$ is rigid. Recall that this makes it possible to define, for every object X , its dimension $\dim(X) \in \mathrm{End}(1) = \mathbb{Q}(t)$ as the composite $1 \rightarrow X^\vee \otimes X \rightarrow 1$ of the evaluation and coevaluation morphisms.
- (iii) The tensor category $\mathcal{C}_t$ is of adjoint type in the following sense. There is an object $\mathfrak{g}$, equipped with a Lie bracket $[\cdot, \cdot] : \mathfrak{g} \otimes \mathfrak{g} \rightarrow \mathfrak{g}$, acting functorially on each object X by an action $\rho_X : \mathfrak{g} \otimes X \rightarrow X$. The action is compatible with tensor products: $\rho_{X \otimes Y} = \rho_X \otimes 1 + 1 \otimes \rho_Y$. For $X = \mathfrak{g}$, ρ_X is $[\cdot, \cdot]$. The action ρ_X transposes to a coaction ${}^t\rho_X : X \rightarrow \mathfrak{g}^\vee \otimes X$. For every X , $X^\natural := \mathrm{Ker}({}^t\rho_X)$ is the largest subobject of X which is a sum of copies of $\mathbf{1}$. Finally, every object is a subquotient of a sum of objects $\mathfrak{g}^{\otimes n}$.
- (iv) $\mathfrak{g}$ is a simple object, and the Killing form is nondegenerate.
- (v) There are simple objects X_i ($0 \leq i \leq 4$), Y_i ($0 \leq i \leq 3$), Y_i^* ($0 \leq i \leq 3$), A , C , and C^* satisfying (A)–(E), with $a = t$.

- (vi) Every simple object S is absolutely simple, i.e. $\text{End}(S) = \mathbb{Q}(t)$, and is isomorphic to its dual; more precisely, it admits a nondegenerate symmetric bilinear form $S \otimes S \rightarrow \mathbf{1}$. The dimension $\dim(S) \in \mathbb{Q}(t)$ of a simple object factors into linear factors.
- (vii) The category $\mathcal{C}_t$ has a $\otimes$ -autoequivalence $*$, semilinear with respect to $t \mapsto -(1/6) - t$, sending A and the X_i to themselves, Y_i to Y_i^* , and C to C^* ; the square of $*$ is isomorphic to the identity.
- (viii) In a sense to be made precise, the category of representations of G is a specialization of $\mathcal{C}_t$ at $t = a$.

A model

Here is a model for this conjecture. Let $\mathcal{B}_n^-[t]$ be the following rigid $\mathbb{Z}[t]$ -linear tensor category.

Objects: finite sets.

$\text{Hom}(X, Y)$: let $\mathcal{C}$ be the set of isomorphism classes of compact one-dimensional manifolds with boundary, whose boundary is $X \amalg Y$; these are bordisms from X to Y . The $\mathbb{Z}[t]$ -module $\text{Hom}(X, Y)$ is the quotient of the free $\mathbb{Z}[t]$ -module $\mathbb{Z}[t]^{(\mathcal{C})}$ with basis $[c]$, $c \in \mathcal{C}$, by the relations

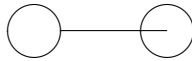
$$[c \cup \text{a circle}] = t[c].$$

Composition: composition of bordisms.

$\otimes$: disjoint union.

For $\lambda \in \mathbb{Q}$, let $\mathcal{B}_\lambda$ be the rigid $\mathbb{Q}$ -linear tensor category obtained from $\mathcal{B}_n^-[t]$ by scalar extension $\mathbb{Z}[t] \rightarrow \mathbb{Q}$, $t \mapsto \lambda$. A morphism $f : X \rightarrow Y$ of $\mathcal{B}_\lambda$ will be called negligible if $\text{Tr}(gf) = 0$ for every $g : Y \rightarrow X$. The negligible morphisms form an ideal I_λ , and if f is negligible then so is $f \otimes g$. Thus the quotient category $\mathcal{B}_\lambda/I_\lambda$ is still rigid tensor. Let $(\mathcal{B}_\lambda/I_\lambda)^{\text{kar}}$ be its Karoubian envelope, obtained by formally adjoining the direct factors corresponding to idempotent endomorphisms.

If λ is an integer $n \geq 0$, this Karoubian envelope is just the category $\text{Rep}(O(n))$ of representations of the orthogonal group $O(n)$. More precisely, let V be a vector space over $\mathbb{Q}$ equipped with a nondegenerate symmetric bilinear form B . Then there is a tensor functor, unique up to unique isomorphism, from $\mathcal{B}_n^-$ to $\text{Rep}(O(V))$, sending the object $\{1\}$ to V and sending the bordism



from $\{1, 2\}$ to $\emptyset$ to $B : V \otimes V \rightarrow \mathbf{1}$. This tensor functor induces an equivalence of $(\mathcal{B}_n/I_n)^{\text{kar}}$ with $\text{Rep}(O(V))$. This is a translation of Weyl's invariant theory for $O(V)$ in $V^{\otimes N}$; see [3], Chapter V, §5.

If λ is an even integer $-2p \leq 0$, the Karoubian envelope $(\mathcal{B}_\lambda/I_\lambda)^{\text{kar}}$ is a variant of the category $\text{Rep}(\text{Sp}(2p))$ of representations of the symplectic group $\text{Sp}(2p)$: the central element $-1 \in \text{Sp}(2p)$ defines a $\mathbb{Z}/(2)$ -grading of each representation V of $\text{Sp}(2p)$, and the commutativity constraint is defined by the Koszul sign rule. Equivalently, in the category of super-representations of $OSp(0|2p)$, one considers the subcategory of subquotients of sums of $V^{\otimes N}$, with V the obvious super-representation.

Similarly, if one considers finite oriented sets S , that is, sets equipped with $\varepsilon : S \rightarrow \{\pm 1\}$, and oriented bordisms, one obtains a rigid $\mathbb{Z}[t]$ -linear tensor category $\mathcal{A}_n^+[t]$. For $\lambda \in \mathbb{Q}$, let $\mathcal{A}_\lambda$ be the rigid $\mathbb{Q}$ -linear category obtained from $\mathcal{A}_n^+[t]$ by the scalar extension $\mathbb{Z}[t] \rightarrow \mathbb{Q}$, $t \mapsto \lambda$. Define the ideal I_λ of negligible morphisms in $\mathcal{A}_\lambda$ as before. If λ is an integer $n \geq 0$, $(\mathcal{A}_\lambda/I_\lambda)^{\text{kar}}$ is just $\text{Rep}(\text{GL}(n))$. If λ is an integer $-n \leq 0$, one obtains the variant of $\text{Rep}(\text{GL}(n))$ defined by $-1 \in \text{GL}(n)$: in the category of super-representations of $\text{GL}(0|n)$, one takes the subcategory of subquotients of sums of $V^{\otimes N} \otimes V^{\vee \otimes M}$, with V the evident super-representation.

In both cases, after extending scalars from $\mathbb{Z}[t]$ to $\mathbb{Q}(t)$, the Karoubian envelope of the category $\mathcal{A}_t$ or $\mathcal{B}_t$ obtained in this way is a semisimple rigid tensor category. Its simple objects S are absolutely simple, and their dimensions $\dim(S) \in \mathbb{Q}(t)$ are polynomials in t that are products of linear factors.

The formulae

Here are the formulae.

Let V_λ^- be the object “one point” of $\mathcal{B}_\lambda^-$, and let V be its image in $\mathcal{B}_\lambda^{-\text{kar}}$, the “evident representation”. To study the subcategory of $\mathcal{B}_\lambda^{-\text{kar}}$ formed by the subquotients of sums of $V^{\otimes i}$, $i \leq N$, one method is to use polynomial continuation from the category $\text{Rep}(O(n))$ for n large. The irreducible objects of $\mathcal{B}_\lambda^{-\text{kar}}$ then correspond to decreasing sequences of integers $\ell_i \geq 0$, with $\ell_i = 0$ for i large enough. The corresponding representation of $O(n)$, for n large, has dominant weight $\sum_i (\ell_i - \ell_{i+1}) \omega_i$ in the notation of the Bourbaki tables. Weyl’s dimension formula gives, by continuation, the dimension of this irreducible object. Let $h((\ell_i))$, or simply h , be the integer such that the representation of the symmetric group defined by the partition (ℓ_i) of $\sum \ell_i$ has dimension $(\sum \ell_i)!/h$. If $\ell_i = 0$ for $i > a$, then

$$h = \prod_{i=1}^a (\ell_i + a - i)! \prod_{1 \leq i < j \leq a} ((\ell_i - i) - (\ell_j - j)).$$

Let $a(n)$, respectively $b(n)$, be the number of pairs i, j , $1 \leq i \leq j$, such that $n = (\ell_i - i) + (\ell_j - j)$, respectively $n = -i - j$. With this notation,

$$\dim V((\ell_i)) = (1/h) \prod (t + n)^{a(n) - b(n)}.$$

Formally, this is

$$(1/h) \frac{\prod (t + \ell_i - i + \ell_j - j)}{\prod (t - i - j)}.$$

It is a polynomial of degree $\sum \ell_i$.

The object $\mathfrak{g} := \wedge^2 V$ plays the role of the adjoint representation: it has a Lie bracket, and acts functorially and compatibly with tensor products on every object. The bilinear form $\text{Tr}(\rho(x)\rho(y), V)$ on $\mathfrak{g}$ is nondegenerate. On $V((\ell_i))$, the corresponding Casimir is

$$\sum_i \ell_i (\ell_i - 2i) + t \sum_i \ell_i,$$

as one checks by continuation from the case of $O(n)$ for n large.

Let V_λ^+ be the object “one positively oriented point” ($\varepsilon = 1$) of $\mathcal{A}_\lambda^+$, and let V be its image in $\mathcal{A}_\lambda^{+\text{kar}}$, the “evident representation”. To study the subcategory of $\mathcal{A}_\lambda^{+\text{kar}}$ formed by the subquotients of sums of $V^{\otimes i} \otimes V^{\vee \otimes j}$, $i, j \leq N$, one method is to use polynomial continuation from the category $\text{Rep}(\text{GL}(n))$ for n large. The irreducible objects of $\mathcal{A}_\lambda^+$ then correspond to pairs of decreasing sequences of positive integers $((\ell_i), (m_i))$, with ℓ_i and m_i zero for i sufficiently large. If the character group of the maximal torus of diagonal matrices in $\text{GL}(n)$ is identified with $\mathbb{Z}^n$ in the usual way, then the corresponding representation of $\text{GL}(n)$, for n large, has dominant weight

$$(\ell_1, \ell_2, \dots, 0, \dots, 0, -m_2, -m_1).$$

The object $\mathfrak{g} := V \otimes V^\vee$ plays the role of the adjoint representation, and the bilinear form $\text{Tr}(\rho(x)\rho(y), V)$ is nondegenerate.

Let $c(n)$, respectively $d(n)$, be the number of pairs i, j , $1 \leq i, j$, such that $n = \ell_i - i + m_j - j + 1$, respectively $n = -i - j + 1$. By continuation of Weyl’s formula,

$$\dim V((\ell_i), (m_i)) = (1/h((\ell_i)) \cdot h((m_i))) \prod (t + n)^{c(n) - d(n)}.$$

Formally, this is

$$(1/h((\ell_i)) \cdot h((m_i))) \frac{\prod (t + \ell_i - i + m_j - j + 1)}{\prod (t - i - j + 1)}.$$

It is a polynomial in t of degree $\sum \ell_i + \sum m_i$. The Casimir relative to $\text{Tr}(\rho(x)\rho(y), V)$ is

$$\sum_i \ell_i(\ell_i - 2i + 1) + \sum_i m_i(m_i - 2i + 1) + t\left(\sum_i \ell_i + \sum_i m_i\right).$$

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This work was inspired by a conversation with P. Vogel, in which he told me that there are three simple Lie groups: linear, orthogonal, and exceptional, depending on parameters. He exploits this point of view in [2], of which numbers 6.8 and 6.9 were especially useful to me.

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